Protein Classification with Deep Reinforcement Learning

2D HP FOLDING · GO PREDICTION · WEB PLATFORM

Arbi Golemi

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ARBI GOLEMI

Introduction

1. Proteins are chains of amino acids that fold into a 3D shape
2. Their 3D shape determines their biological function
3. Misfolded proteins cause Alzheimer's, Parkinson's, and cancer
4. Millions of sequences known (UniProt) → far fewer solved structures (PDB)
5. Even a simplified model of folding is NP-complete to solve exactly

Project Goal

1. Classical heuristic algorithms (HC, SA, MC, REMC)
2. Tabular Q-Learning
3. Deep Q-Network (DQN)
4. GO term prediction via DeepGO API
5. 3D structure retrieval (AlphaFold / ESMFold)

The Hydrophobic-Polar (HP) Model

- Simplifies 20 amino acids into 2 types:
  - H (Hydrophobic):
  - P (Polar)
- Protein placed on a 2D square grid as a self-avoiding walk
- Energy = −(number of non-consecutive H-H contacts)
- Goal: minimize energy (more contacts = more stable protein)

Neighbour Definition

Two residues i and j are topological neighbours if:

- They are adjacent on the grid (Manhattan distance = 1)
- They are NOT consecutive in the sequence ($|i - j| > 1$)

Exploring the Conformation Space

All 6 algorithms use the same 4 local moves

Move	Description
End-flip	Move the first or last residue to any free adjacent site
Kink-jump	If residue i forms a 90° bend → flip it to the opposite corner
Crankshaft	Rotate a U-shaped pair of residues 180°
Pivot	Rotate all residues after a pivot point by ±90° or 180°

Hill Climbing (HC)

Pros:
Very fast (under 1 second for 10,000 steps)
Simple and deterministic

Cons:
Easily trapped in local minima
Cannot escape once stuck

- Start from a straight-line conformation
- Pick a random move at each step
- Accept only if energy improves or stays the same: $E(new) \leqslant E(current)$
- Keep track of the global best solution

Simulated Annealing

- Inspired by annealing: Temperature decreases
- Accepts bad moves early ($P = e^{(-\Delta E/T)}$)
- High T: Exploration | Low T: Exploitation
- Cooling: T from 10.0 to 0.01

Monte Carlo Methods

Monte Carlo (MC)

- Same Metropolis acceptance rule as SA
- Temperature stays constant
- Good for sampling conformations at a fixed temperature
- Performs worse than SA/HC at energy minimization

Replica Exchange MC (REMC)

- Runs 3 parallel replicas at temperatures 1.0 / 2.0 / 5.0
- Every 100 steps: adjacent replicas try to swap conformations
- Low-temperature replicas benefit from high-temperature exploration
- Result: comparable quality to SA, but ~8× slower

Reinforcement Learning

Component	Definition
State (S)	Current conformation of the protein on the grid
Action (A)	One of 4 moves: end-flip, kink-jump, crankshaft, pivot
Reward (R)	E(old) - E(new) if valid move; −1 if invalid (collision)
Discount (γ)	0.95 — balances immediate vs future reward

Tabular Q-Learning

- Maintains a Q-table: Q(state, action) = expected future reward
- Bellman update at each step
- Learning rate $\alpha = 0.1$, discount $\gamma = 0.95$
- explore randomly (ε goes from $1.0 \rightarrow 0.05$ during training)
- State: normalized conformation tuple

Results

- Effective for sequences up to ~25 amino acids
- State space grows exponentially with sequence length

Deep Q-Network (DQN)

Architecture:

Input: 200×200 grid
3 Convolutional blocks
Flatten → Linear(40,000 → 256) → Linear(256 → 4 actions)

Training

Experience Replay
Target Network: slow-updating copy for stable training
Loss: Mean Squared Error on Q-values
Optimizer: Adam (learning rate = 0.001)

Inference strategy

Phase 1 (ε = 0.4): DRL-guided exploration
Phase 2 (ε = 0): Greedy local search from the best structure found in Phase 1

Web Platform Architecture

- Framework: Django (Python)
- Frontend: HTML5 + CSS3 + Bootstrap 5
- Visualization: SVG auto-scaling lattice
- Architecture: Model-View-Template (MVT)

APIs

- DeepGO: Predict GO terms (Biological Process, Molecular Function, Cellular Component)
- AlphaFold DB: Fetch 3D structure by UniProt accession code

Web Platform UI

- FASTA text area (supports multiple sequences)
- Action selector: GO Prediction, 2D Structure, 3D Structure
- Algorithm selection
- Dynamic hyperparameter panel (iterations, temperature, episodes, learning rate)

- 2D structure map viewer H/P string, energy and parameters displayed
- GO terms with confidence scores, grouped by ontology domain
- 3D structure fetch by UniProt accession

Benchmark Testing Setup

Dataset: 20 real UniProt protein sequences

Sequence length: 56 – 98 amino acids

Iteration budgets: 1,000 / 10,000 / 50,000 / 100,000

Repetitions: 10 runs per configuration (best + average reported)

Metric: HP energy (lower = better)

Results at 50,000 Iterations

Protein	Length	H%	HC	SA	MC
A1L190	88 AA	41%	−25	−25	−13
O00168	92 AA	42%	−25	−28	−19
O95777	96 AA	43%	−29	−29	−14
P04155	84 AA	46%	−26	−28	−19
P06028	91 AA	53%	−30	−31	−22
P29034	98 AA	43%	−27	−30	−19

Results Q-Learning & DQN

Tabular Q-Learning (example: O00168, 92 AA)

Steps	Best Energy
1,000	−7
10,000	−8
50,000	−10
100,000	−11

DQN results

Protein	Length	DQN Energy	Training Time
A1L190	88 AA	−3	~38 minutes
C9JLW8	97 AA	−1	~47 minutes

Conclusions

- Full Django web platform 6 algorithms, GO prediction, 3D retrieval, 2D structure
- SA > HC > MC in solution quality at same computation budget
- Q-Learning limited by curse of dimensionality above ~25 AA
- DQN demonstrates spatial CNNs can learn protein folding patterns

Future Directions

DQN at scale

GPU training, 100k–1M episodes, Bayesian hyperparameter search

Advanced DRL

Explore PPO or SAC for better sample efficiency

Database

Train with more proteins

Idea

Integrate DRL with classical algorithms for better performance

The End

THANK YOU FOR LISTENING

Arbi Golemi